

THENMOZHI SEKAR

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About Me

AI/ML Engineer with 5 years 11 months of experience in Machine Learning, Generative AI, Large Language Models (LLMs), and scalable AI systems. Experienced in developing production-grade AI applications, LLM-powered solutions, NLP systems, and end-to-end ML pipelines using Python, PyTorch, AWS, and modern MLOps practices. Published multiple research papers in IEEE venues, demonstrating expertise in applied AI, machine learning, and intelligent software systems.

EDUCATION

Quinnipiac University,
Master of Science in Computer Science, CGPA – 3.8/4
Madras University,
Bachelor of Computer Applications, CGPA - 3.7/4

Hamden, CT, USA
Aug 2024 – May 2026
Kanchipuram, TN, India
Aug 2018 – May 2021

EXPERIENCE

Graduate Research Assistant - AI, Machine Learning & Software Engineering (Part-time)
Quinnipiac University, Hamden, CT, USA

Aug 2024 – Present

- Designed and implemented AI/ML pipelines involving NLP, document understanding, feature engineering, LLM workflows, and experimental analysis using Python and PyTorch
- Conducted AI and Software Engineering research focused on intelligent systems, semantic analysis, AI-generated code evaluation, and machine learning-driven applications
- Collaborated with faculty advisors on experimentation, model evaluation, prompt engineering, technical documentation, and publishable AI/ML research projects

Machine Learning Engineer (Remote)
Capital One, New York, United States

May 2025 – Dec 2025

- Designed and deployed ML models for fraud detection and credit risk analysis, improving prediction accuracy by **18%** using **Python, SQL, and Scikit-learn**.
- Built end-to-end ML pipelines processing **5M+** financial records, reducing training time by **30%** using Spark and feature engineering.
- Deployed ML models on **AWS** using **FastAPI, Docker, and CI/CD**, reducing inference latency by **25%**.

Applied AI Engineer (Full Time)
Uniphore, Chennai, India

Jun 2022 – Jul 2024

- Developed and deployed AI-powered applications using Large Language Models (LLMs), AI agents, NLP, and machine learning techniques, automating enterprise workflows and improving operational efficiency by **30%**.
- Built scalable AI agents and RAG pipelines, reducing response and information retrieval times by **40%**.
- Collaborated with cross-functional teams to optimize model performance and production deployments, improving AI system reliability by **25%**.

Full-Stack Java Software Engineer (Full Time)
Ijona Technologies Pvt. Ltd, Bangalore, India

Apr 2021 – May 2022

- Developed and validated backend algorithmic workflows using Java and SQL to improve software scalability, reliability, and operational efficiency.
- Conducted user behavior and system interaction analysis to optimize application design, document processing workflows, and system performance.
- Improved system reliability through debugging, testing, performance analysis, and continuous optimization across software engineering workflows.

PROJECTS

LLM-Based Bug Report Enhancement Pipeline | Python, NLP, LLMs

Jan 2026 – Present

- Built an LLM-based NLP pipeline to transform unstructured bug reports into standardized formats, improving report completeness by 72% across 15,000+ GitHub API bug reports.
- Developed automated preprocessing and structured information extraction workflows to improve debugging, issue triaging, and semantic analysis.

AI-Based Code Generation Evaluation | Python, Java, Published Research

Aug 2025 – Feb 2026

- Conducted empirical evaluation of AI-generated code using 200+ code samples and 1200+ experimental runs across similarity and logical correctness metrics.
- Published research at ISEC 2026 focused on assessing AI-assisted software engineering and AI-generated code quality evaluation techniques.

Perceived Stress Prediction Using Machine Learning with Feature Selection | Python, XGBoost, Scikit-learn

Sep 2025 – May 2026

- Developed machine learning models to predict perceived stress levels using EMA stress and sleep datasets from the StudentLife research study.
- Applied feature engineering, mutual information-based feature selection, and cross-validation using XGBoost, SVM, Random Forest, and Neural Networks.

Flow Pilot: Enterprise AI Workflow Automation Platform | Next.js, TypeScript, PostgreSQL

Jan 2026 – May 2026

- Built a full-stack enterprise AI workflow automation platform supporting workflow orchestration, approval systems, and API-driven operations.
- Developed scalable backend workflows and database-integrated features using Next.js, TypeScript, PostgreSQL, Prisma, and modern engineering practices.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, JavaScript, TypeScript, C/C++

Machine Learning & AI: Machine Learning, Deep Learning, PyTorch, Scikit-learn, XGBoost, NLP, Transformers, Hugging Face, Feature Engineering, Model Training, Model Validation, Hyperparameter Tuning

Generative AI: Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, LangChain, Semantic Search, OpenAI API, LLM Workflows, Vector Databases

MLOps & Cloud: AWS, Docker, Kubernetes, MLflow, Airflow, CI/CD, Git, Model Deployment, Model Monitoring, Experiment Tracking, GitHub Actions

Backend & Frameworks: FastAPI, REST APIs, Next.js, Node.js, Prisma, PostgreSQL

Data Engineering & Analytics: SQL, PostgreSQL, Pandas, NumPy, Apache Spark, Data Preprocessing, Data Analysis

Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, System Design, SDLC, Agile, Microservices, Scalable Distributed Systems

Developer Tools: GitHub, VS Code, IntelliJ IDEA, Jupyter Notebook, Postman

PUBLICATIONS

- Sekar, T., & ElKharchoutly, R., “Assessing AI-Based Code Generation Using Code Comments,” 2026 5th International Conference on Information Systems Engineering (IISEC), IEEE Xplore, 2026
 - Sekar, T., ElKharchoutly, R., & Sahoo, S., “Perceived Stress Prediction Using Machine Learning with Feature Selection on EMA Stress and Sleep Data: A StudentLife Study,” at IEEE/ACM CHASE 2026 Poster & Demo Track
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